



Dynamic, real-time HVAC optimisation  
- delivering energy savings automatically.

# Local Government Civic Office

Dynamic, real-time HVAC optimisation case study

|                                                                                                                                             |                                                                                                 |                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------|
| <div>£37.5k</div> <div>138 MWh</div> <div>8,500 m²</div>                                                                                    |                                                                                                 |                                       |
| Calculated energy cost saving<br>first 12 months                                                                                            | First-year energy reduction<br>against expected profile                                         | Civic office<br>existing BMS retained |
| Assessed against the expected post-decarbonisation energy profile   Wider low-carbon measures already underway   Existing controls retained |                                                                                                 |                                       |
| 17.9%                                                                                                                                       | 8.0%                                                                                            |                                       |
| Electricity below expected consumption<br>222,278 kWh lower than the expected first-year profile                                            | Total energy below expected consumption<br>137,701 kWh net reduction across electricity and gas |                                       |

## Executive summary

A UK local authority was implementing a wider building decarbonisation programme across its estate, including heat pumps, solar generation, battery storage and Building Management System improvements. At an 8,500 m² civic office, EcoAI's HVAC optimisation technology was integrated with the existing BMS to improve day-to-day operation while retaining the building's installed controls infrastructure.

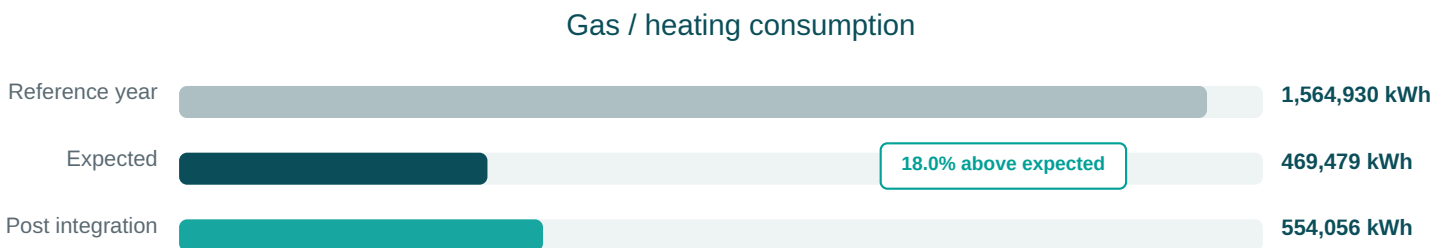
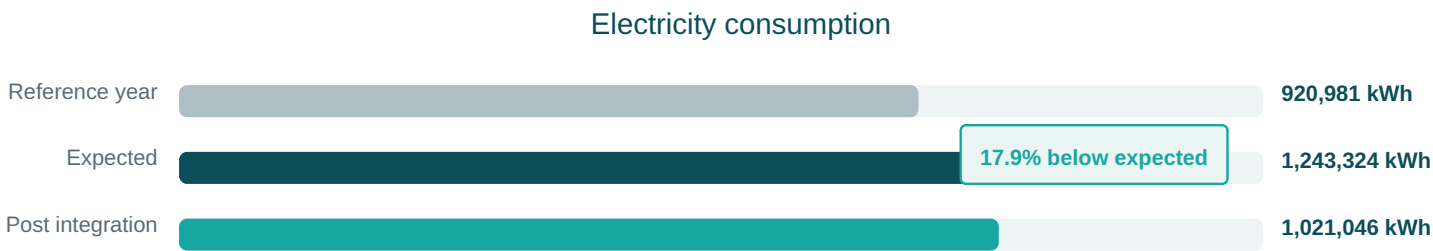
The challenge was not simply to compare one year with another. The wider carbon-reduction measures were expected to materially change the building's energy profile, with electricity consumption forecast to increase by 35% and gas consumption forecast to reduce by 70%. Performance was therefore assessed against the expected post-project profile.

Across the first 12 months, electricity consumption was 222,278 kWh below expected. Gas consumption was 84,577 kWh above the expected level, giving a net energy reduction of 137,701 kWh. At the rates used in the verification, this equated to a calculated net saving of £37,474.87.

The project demonstrates how continuous HVAC optimisation can work alongside wider decarbonisation investment, improving operational performance and extracting more value from existing controls infrastructure without requiring it to be replaced.

## First-year performance against expected profile

The first-year verification compared actual post-integration performance with the energy profile expected after the site's wider decarbonisation measures. This is important because the building was undergoing a major shift from gas towards electricity, making a simple historic year-on-year comparison potentially misleading.



Gas use remained c.65% below the original reference year, but 18.0% above the reduced level expected from wider measures. Even so, EcoAI's electricity savings delivered a 137,701 kWh net energy reduction and £37,474.87 calculated energy cost saving.

Expected profile: the wider project anticipated a 35% increase in electricity use and a 70% reduction in gas use as other carbon-reduction measures were implemented.

| Result                        | First-year position                                                   |
|-------------------------------|-----------------------------------------------------------------------|
| Electricity                   | Expected: 1,243,324 kWh   Actual: 1,021,046 kWh   Saving: 222,278 kWh |
| Gas / heating                 | Expected: 469,479 kWh   Actual: 554,056 kWh   Variance: +84,577 kWh   |
| Net energy position           | 137,701 kWh reduction against expected consumption                    |
| Electricity cost              | Expected: £246,802.80   Actual: £202,680.07   Saving: £44,122.73      |
| Gas / heating cost            | Expected: £36,901.65   Actual: £43,549.51   Variance: +£6,647.86      |
| Calculated energy cost saving | £37,474.87                                                            |

## Background and deployment

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The local authority had invested significantly in energy-efficiency and carbon-reduction measures across a portfolio of public buildings. The civic office combined an existing BMS with a range of mechanical systems and was selected as a practical location to improve operational performance alongside the wider programme.

EcoAI's optimisation technology was integrated with the existing Building Management System, using live building conditions together with local weather and forecast information to improve heating, ventilation and cooling operation while making greater use of the building's thermal behaviour.

The existing BMS and plant remained in place. During operation, the project also helped identify systems requiring attention, providing useful operational insight for the engineering team alongside the energy optimisation.

## Working alongside decarbonisation

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This project is particularly relevant to local government because the building's energy profile was changing as other measures were introduced. Heat electrification and other carbon-reduction projects can shift demand between fuels, which means optimisation performance should be assessed against the expected future operating profile, not just a historic baseline.

Actual gas use remained around 65% below the original reference year, but 18% above the reduced level anticipated by the wider programme. Even so, EcoAI's electricity savings were sufficient to deliver a 137,701 kWh net energy reduction and £37,474.87 calculated energy cost saving against the expected profile. This shows how continuous HVAC optimisation can continue to add measurable value alongside wider capital decarbonisation measures.

## First-year verification approach

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The facilities energy team reviewed historical consumption, the expected consumption after the wider efficiency measures, and the first 12 months following integration. Changes in unit rates and the expected shift in fuel use were incorporated so the reported financial and energy position reflected the optimisation project rather than simply comparing raw annual totals.

Verification note. The figures in this case study reproduce the first-year project analysis. The calculated £37.5k energy cost saving is the difference between the expected and actual energy costs used in that analysis.

## What the project demonstrates

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### EXISTING BMS

Optimisation can work through existing controls infrastructure rather than requiring a wholesale replacement project.

### WIDER PROGRAMME

Operational optimisation can complement heat pumps, on-site generation, storage and other decarbonisation measures.

### MEASURED VALUE

Performance can be evaluated against an adjusted expected profile when other projects are changing the building's energy mix.

**A practical local-government reference for optimising existing buildings as estates decarbonise.**